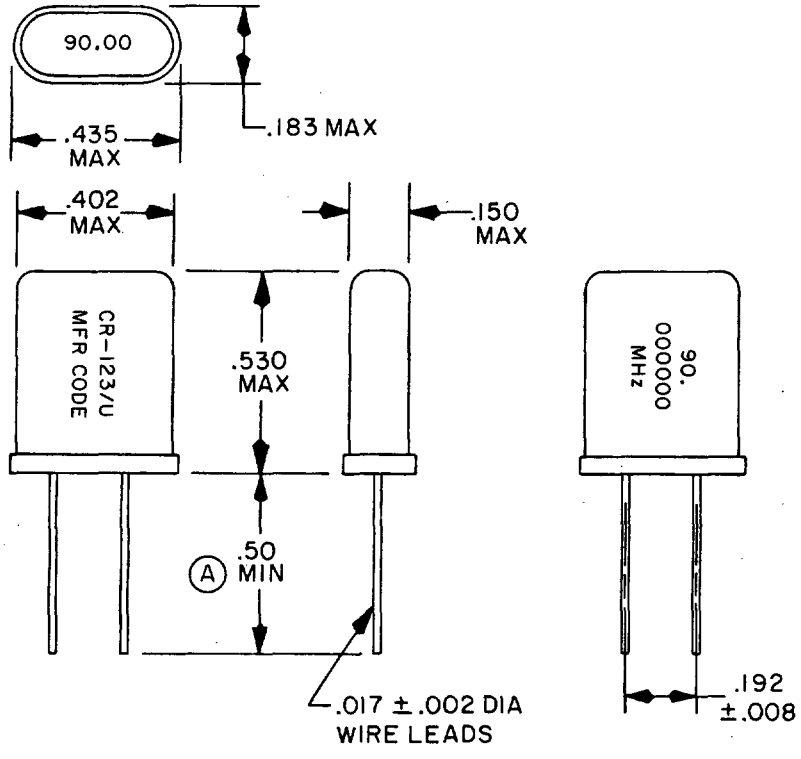
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-123/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; controlled; series-resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-123/U.

Ⓐ denotes changes

FSC 5955

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Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	50 to 125 MHz
Capacitance, shunt	4.5 pF, maximum
Frequency tolerances:	
Operating temperature range	±10 ppm
Reference temperature	±10 ppm
Room temperature	±80 ppm
Frequency stability	±2.5 ppm
Mode of oscillation	Fifth mechanical overtone
Government reference standard test set(s)	Up to 100 MHz TS-683/TSM Above 100 MHz AN/TSM-15
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Resonance	Series
Reference temperature	85°C ±1°C
Temperature ranges:	
Operating (controlled)	80°C to 90°C, inclusive
Operable	-55°C to +80°C and +90°C to +100°C
Equivalent resistance	60 ohms, maximum
Unwanted modes	Minimum unwanted mode effective resistance of 120 ohms or an unwanted-to-main mode resistance ratio of 3:1, whichever is greater.
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock	±10 ppm
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A, 0.06 inch double amplitude or 5 gravity units, whichever is less.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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